

THE ZERO PARADOX

ZP-J: Executability of Self-Reference

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This document establishes Theorem T-EXEC (Executability of Self-Reference): in any ZP-A join-semilattice carrying the AFAStructure typeclass, whatever fills the Quine-atom role — the lattice-level encoding of the set $Q = \{Q\}$ of ZF+AFA (Anti-Foundation Axiom) — is provably the bottom element $\perp$. This bridges the set-theoretic and order-theoretic layers of the framework at the level of that role; the set identity itself stays in ZF+AFA. CC-1 from ZP-A, which stated " $S_0 = \perp$ " as a modelling commitment, is restated rather than forced: given the typeclass, starting at a Quine atom and starting at $\perp$ are the same condition.

This layer extends the core T-EXEC result in four directions: it identifies the set of self-containing elements as $\{\perp\}$ with no Classical.choice, and states why that does not bear on Aczel's question about Dependent Choice (Section VII); it generalises the AFAStructure typeclass into an abstraction chain reaching down to a pure valuation structure (Section VIII); it instantiates that chain on two concrete types (Section IX); and it proves global decoration uniqueness, into abstract DecorationUniverses, for finite accessible pointed graphs (Section X). All results are sorry-free in Lean 4.

Section I: The Open Question — CC-1 as a Modelling Commitment

I. CC-1 in ZP-A

ZP-A Conditional Claim CC-1 states: if the state sequence is initialised at $\perp$, then $\perp \leq S(n)$ for all n . This follows trivially from T2 ($\perp$ is the global minimum), so its formal content is essentially: we are choosing $\perp$ as the initial state S_0 .

The label "Conditional Claim" (CC) marks this as a modelling commitment — an explicit choice not derivable from the axioms A1–A4 alone. ZP-A's axioms give us a semilattice with a bottom element. They do not say which instantiation of the semilattice should start at that bottom. CC-1 asserts: ours does. This is well-motivated, but it is a choice.

The question ZP-J investigates: is this choice forced? Is there a structural reason — derivable from the framework's foundational commitments — that any well-grounded instantiation of ZP-A must begin at $\perp$? The answer is that the choice is restated, not forced: if the lattice carries AFAStructure, a sequence starts at a Quine atom exactly when it starts at $\perp$ (Section V). The commitment to start there remains; it is now expressed through the Quine-atom role rather than as a bare choice of element.

II. The Implicit Identification

ZP-E's DA-1 Path 1 already states: $\perp = \{\perp\}$ (Quine atom, ZF+AFA). This identification — the bottom element is the unique self-containing set under AFA — was present informally but never formally bridged to CC-1. It appeared as motivation for why $\perp$ is the right starting point, not as a derivation of it.

The gap: ZP-A's lattice order is abstract (defined by axioms A1–A4). AFA is a set-theoretic axiom. There is no automatic connection between "x is self-containing" and "x is the lattice bottom." Connecting them requires a bridge — and that bridge was the missing piece. ZP-J provides it.

Open question entering ZP-J: Is CC-1 ($S_0 = \perp$) forced by the framework's foundational structure, or is it an independent modelling choice? If forced, what is the structural reason? Answer (ZP-J T-EXEC): not forced, restated. Given an AFAStructure, "starts at a Quine atom" and "starts at $\perp$ " are equivalent (`t_exec_iff`), so the starting point is named by its role; the choice to start there remains. The AFA identification $\perp = \{\perp\}$ is the ZF+AFA reading of the Quine-atom role the typeclass encodes; the Lean states the role, not the set identity.

Section II: AFA Machinery — Self-Membership and the Quine Atom

I. The Anti-Foundation Axiom

Standard set theory (ZF) includes the Foundation Axiom: every non-empty set S contains an element disjoint from S . A consequence is that no set can contain itself: $x \in x$ is impossible in ZF. This rules out self-referential sets by fiat.

The Anti-Foundation Axiom (AFA, Aczel 1988, ch. 1 p. 6) replaces Foundation with an axiom asserting existence and uniqueness: every graph has a unique decoration. As a consequence, every accessible pointed graph (APG) is a picture of a unique set. Under AFA, self-containing sets are not only possible but uniquely characterised. The resulting theory ZF+AFA is consistent and expressive — it is the natural set-theoretic home for fixed-point and self-referential structures.

II. The Quine Atom

Under AFA, the equation $x = \{x\}$ has a unique solution. This solution is called the Quine atom, denoted Q . Q is the unique set that contains itself as its sole member: $Q = \{Q\}$. It is not the empty set ($\emptyset = \{\}$ contains nothing); it is not any well-founded set (those cannot contain themselves). Q is the minimal non-trivial AFA set — it contains exactly one thing, and that thing is itself.

The AFA uniqueness guarantee is the key property: there is at most one Quine atom. Any two self-containing elements are equal. This means the self-membership property uniquely identifies an element — a fingerprint that belongs to exactly one object in the universe.

AFA Uniqueness (`AFAStructure.quine_unique`)

For any type L with AFA structure: if $x, y \in L$ both satisfy `selfMem(x)` and `selfMem(y)`, then $x = y$.

Informally: the Quine atom is unique. Self-containment is a property held by at most one element of any AFA-structured type.

Lean: `AFAStructure.quine_unique` — encoded as a typeclass field, not a theorem, because AFA is a foundational axiom and cannot be derived from type-theoretic principles alone for an arbitrary `selfMem`. It is a hypothesis every theorem here carries, and (Section III.II) any lattice can meet it by a trivial choice of `selfMem`.

III. Self-Membership as a Lattice Predicate

In ZP-J, the AFA machinery is abstracted minimally. We do not need the full apparatus of accessible pointed graphs, bisimulation, or set decoration. We need only two things: a predicate `selfMem` on the lattice elements, and the guarantee that it is held by at most one element (`quine_unique`). The third structural field — `bot_self_mem` — is where the bridge is built.

The `AFAStructure` typeclass in `SetTheoryAFA.lean` captures exactly this. Its three fields are sufficient to derive T-EXEC with no additional axioms. The full AFA machinery (APGs, bisimulation, decoration) motivates the fields; the formal derivation requires only the fields themselves, and every `ZPSemilattice` supplies them (Section III.II).

Section III: AFAStructure — The Structural Bridge

I. The Three Fields

The `AFAStructure` typeclass for a ZP-A semilattice L has three fields. The first two encode standard AFA properties. The third is the bridge.

AFAStructure Typeclass (SetTheoryAFA.lean § I)

class `AFAStructure` ($L : \text{Type}^*$) [`ZPSemilattice` L] with:

(1) `selfMem` : $L \rightarrow \text{Prop}$ — the self-containment predicate. Read in ZF+AFA, `selfMem(x)` is $x = \{x\}$ (x is its own sole member), not bare membership $x \in x$: under AFA the set $0^* = \{\emptyset, 0^*\}$ contains itself and is not Q (Aczel 1988, Example 1.5), so the uniqueness below would fail for membership.

(2) `quine_unique` : $\forall x y : L, \text{selfMem}(x) \rightarrow \text{selfMem}(y) \rightarrow x = y$ — AFA uniqueness. At most one element of L plays the self-containment role.

(3) `bot_self_mem` : `selfMem`($\perp$) — the bridge field. The bottom element of the lattice is self-containing. This encodes the ROLE that $\perp = \{\perp\}$ describes in ZF+AFA; the set identity itself compares an element of L with a set of elements of L , so it is not stated here.

II. What bot_self_mem Says

`bot_self_mem` is the single structural claim that connects the order-theoretic world (ZP-A's lattice) to the set-theoretic world (AFA). It says: the bottom element $\perp$ of the lattice is self-containing — it satisfies `selfMem`($\perp$).

In set-theoretic terms, within ZF+AFA: $\perp = \{\perp\}$, so in particular $\perp \in \perp$. This is the identification that ZP-E's DA-1 Path 1 invokes informally. ZP-J encodes the role it describes as a typeclass field, so every result that uses it states it as a hypothesis rather than leaving it as narrative motivation.

Supplying the field is not a test a lattice can fail. Every `ZPSemilattice` carries an `AFAStructure`: take the constant-bottom self-application `trivialSelfApp` (`abstractSelfApp_always_inhabited`, `ZeroParadox/Computability/SelfApp.lean` § V) and pass it through the instance `toAFAStructure`, so that `selfMem(x)` holds exactly when $x = \perp$. Carrying the structure therefore says nothing about a particular lattice. The class encodes a ROLE, and T-EXEC identifies that role's occupant. The `selfMem` field is itself

forced in every instance: `selfMem_determines_singleton` (Section VII.III) gives $\{x : L \mid \text{selfMem}(x)\} = \{\perp\}$ for any `AFAStructure` lattice `L`. Where a concrete instance carries content, it is in the map `selfMem` is defined from (a self-application or a scale, Section VIII.IV), never in `selfMem`.

Remark R-J.0 — CIC Encoding and the AFA Distinction

In the `MachinePhase` concrete instance (ZP-K), `selfMem` is defined as `selfMem x := x = ⊥`. With this definition, `bot_self_mem` is proved by `rfl`: $\perp = \perp$ holds by reflexivity.

This is the CIC-compatible encoding of AFA self-containment. In Lean 4 (based on the Calculus of Inductive Constructions), the set-theoretic statement $\perp \in \perp$ — meaning $\perp$ literally contains itself as a member under `ZF+AFA` — is not well-typed on a carrier: $x : L$ is a member of a Set `L`, and no membership relation between two elements of `L` is in scope, so $\perp \in \perp$ does not elaborate. Mathlib's own model of ZF sets satisfies Foundation (`ZFSet.mem_irrefl`: no set is a member of itself). The encoding `selfMem x := x = ⊥` captures the structural role: "self-containing" means "equals the bottom element." The Lean proof compiles by `rfl` because self-containing is defined as equality with $\perp$.

This is a structural analogy, not a set-theoretic derivation from `ZF+AFA`. The full set-theoretic content of AFA — that $\perp$ literally contains itself as a member of the AFA universe — is not present in the Lean proof. What the typeclass encodes is the structural consequence: there is a unique element playing the Quine role, and $\perp$ is that element.

III. Why a Typeclass Field Rather than a Freestanding Axiom

In the stub version of `SetTheoryAFA.lean`, the bridge was a freestanding axiom: `ax_j1_quine_join_identity`. It stated directly that the Quine atom satisfies the join-identity. This compiled, but the purity check showed T-EXEC depending on `ax_j1_quine_join_identity` — a named axiom floating outside any typeclass.

A freestanding axiom is a global assertion the kernel accepts without proof, and it appears in the purity report of every theorem that uses it. A typeclass field adds nothing to that report: T-EXEC depends on no axioms. What the field buys is honesty of scope, not content — it is a hypothesis each theorem carries, and (Section III.II) every lattice can supply it trivially.

The distinction: a freestanding axiom says "trust me, this is true," and the purity check records it. A typeclass field says "given this structure," and every theorem is conditional on it. ZP-J uses the second. The literal set identity ($\perp = \{\perp\}$) stays in `ZF+AFA`; the Lean carries only the role it describes.

Section IV: Theorem T-EXEC — The Quine Atom is Bottom

Theorem T-EXEC — Executability of Self-Reference

Statement: Let `L` be a ZP-A semilattice with `AFAStructure`. If `q : L` is a Quine atom (`selfMem(q)` and `q` is unique among self-containing elements), then `q = ⊥`.

Lean: `ZeroParadox.t_exec` — proved in `SetTheoryAFA.lean`. Purity: does not depend on any axioms. ✓

I. The Proof

The proof of T-EXEC is immediate from the typeclass fields. It has three steps:

- `hq.2` states: every self-containing element of L equals q . (This is the uniqueness half of `IsQuineAtom q`.)
- `AFAStructure.bot_self_mem` states: $\perp$ is self-containing. (This is the bridge field — the role $\perp = \{\perp\}$ describes in $ZF+AFA$.)
- Applying `hq.2` to $\perp$ using `bot_self_mem` gives: $\perp = q$. By symmetry: $q = \perp$. QED.

The entire proof is one line in Lean 4: `(hq.2 bot AFAStructure.bot_self_mem).symm`. No appeal to the join operation. No bridge axiom. No `DA-2`. Just AFA uniqueness applied at $\perp$.

Remark R-J.1

The proof does not use `da2_bottom_characterization` (from `ZP-E`). That result — “ $(\forall x, \text{join } S \ x = x) \leftrightarrow S = \perp$ ” — is used in the derived theorem `J1` (Section V), but `T-EXEC` itself uses neither the order nor the join: it uses only `bot_self_mem` and the uniqueness half of its own hypothesis `IsQuineAtom q` (not the `quine_unique` field).

II. IsQuineAtom bot

A corollary of `T-EXEC` is that $\perp$ itself is a Quine atom. This is the converse direction: not only does the Quine atom equal $\perp$ (`T-EXEC`), but $\perp$ equals the Quine atom (`bot_is_quine_atom`).

Proposition — bot_is_quine_atom

In any `AFAStructure` lattice L : `IsQuineAtom( $\perp$ )`.

Proof: $\perp$ is self-containing by `bot_self_mem`. Any other self-containing x satisfies $x = \perp$ by `quine_unique(x,  $\perp$ , selfMem(x), bot_self_mem)`.

Lean: `ZeroParadox.bot_is_quine_atom` — does not depend on any axioms. ✓

III. The Full Biconditional

Combining `T-EXEC` and `bot_is_quine_atom` yields the full biconditional: `IsQuineAtom(q)  $\leftrightarrow$  q =  $\perp$` . Being the Quine atom and being the bottom element are the same property, stated as a self-containment role (the lattice encoding of the $ZF+AFA$ set $Q = \{Q\}$) and in order-theoretic language respectively.

Theorems t_exec_iff and t_exec_triple_iff — Full Equivalence

For any $q : L$ in an `AFAStructure` lattice:

`t_exec_iff`: `IsQuineAtom(q)  $\leftrightarrow$  q =  $\perp$` .

`t_exec_triple_iff`: `IsQuineAtom(q)  $\leftrightarrow$  (q =  $\perp$   $\wedge$   $\forall x : L, \text{join } q \ x = x)$` .

The three conditions are mutually equivalent: Quine atom (the self-containment role, encoding the set-theoretic $Q = \{Q\}$), bottom element (order-theoretic), and join-identity element (algebraic). They are three formulations of one structural role.

Lean: `ZeroParadox.t_exec_iff`, `ZeroParadox.t_exec_triple_iff` — do not depend on any axioms. ✓

Section V: Derived Results — J1, and CC-1 Restated

I. J1 — QuineJoinIdentity (Derived)

In the initial stub of SetTheoryAFA.lean, the claim "the Quine atom satisfies the join-identity" was stated as a freestanding axiom (ax_j1_quine_join_identity). ZP-J replaces it with a theorem.

Theorem J1 — QuineJoinIdentity (formerly Axiom AX-J1)

In any AFAStructure lattice L , if q is the Quine atom, then $\forall x : L, \text{join } q \ x = x$.

Proof: $q = \perp$ by T-EXEC. Then $\text{join } q \ x = \text{join } \perp \ x = x$ by A4 (bot_join, ZP-A). ✓

Status: DERIVED THEOREM. Was axiom ax_j1_quine_join_identity in the stub — now proved from T-EXEC + ZP-A A4. No freestanding axiom remains.

Lean: ZeroParadox.j1_quine_join_identity — does not depend on any axioms. ✓

II. CC-1 Restated

cc1_derived proves: if the initial state S_0 of a state sequence is a Quine atom Q , then $S_0 = \perp$. By t_exec_iff the converse holds as well, since $\perp$ is a Quine atom. So "starts at a Quine atom" and "starts at $\perp$ " are the same condition, and CC-1's choice of starting point is restated in terms of the Quine-atom role, not eliminated.

Theorem CC-1 (Derived) — cc1_derived

Let L be an AFAStructure lattice. Let $S : \mathbb{N} \rightarrow L$ be a state sequence (ZP-A D3) and $Q : L$ a Quine atom. If $S(0) = Q$, then $S(0) = \perp$.

Proof: $S(0) = Q$ (hypothesis). $Q = \perp$ by T-EXEC. Therefore $S(0) = \perp$.

Starting at Q and starting at $\perp$ are not two choices — with t_exec_iff they are the same condition. The commitment to start there is not removed; it is stated as starting in the Quine-atom role.

Lean: ZeroParadox.cc1_derived — does not depend on any axioms. ✓

III. Uniqueness of the Bottom Role

A further consequence is algebraic uniqueness: in any ZP-A semilattice (without even requiring AFA structure), at most one element can satisfy the join-identity. This is a clean corollary of da2_bottom_characterization (ZP-E).

Theorem — bot_unique

For any ZP-A semilattice L (no AFA required): if $x, y : L$ both satisfy $\forall z, \text{join } x \ z = z$ and $\forall z, \text{join } y \ z = z$, then $x = y$.

Proof: da2_bottom_characterization gives $x = \perp$ and $y = \perp$. Therefore $x = y$.

Lean: ZeroParadox.bot_unique — does not depend on any axioms. ✓

Section VI: Implications — What Was the Commitment?

I. The Remaining Foundation

T-EXEC and its corollaries carry zero freestanding axioms in the Lean 4 purity check. The entire derivation traces to the fields of two typeclasses: ZPSemilattice (from ZP-A) and AFAStructure (new in ZP-J). No standalone axiom statement appears anywhere in the SetTheoryAFA.lean proof obligations.

The foundational commitments are the typeclass fields themselves. In ZPSemilattice: A1–A4 (the semilattice axioms). In AFAStructure: selfMem (a predicate), quine_unique (AFA uniqueness), and bot_self_mem (the bridge). They are not axioms floating in the ambient theory — they are hypotheses every theorem here carries. They are not facts about any particular lattice: every ZPSemilattice supplies them trivially (Section III.II).

II. Where the Philosophy Lives Now

The philosophical content of $\perp = \{\perp\}$ is not in the Lean. It lives in ZP-A as CC-2 (Self-Containment of $\perp$), a Forced Metatheoretic Commitment with two halves: that any host of a self-membered bottom is not well-founded is proved (quineHost_not_wellFounded, zfSet_no_quine_bottom, ZeroParadox/Settheory/QuineHost.lean), and that a unique Quine atom is the right requirement is argued. It lives too in ZP-E's DA-1 Path 1 (one of three informal arguments for instantiation as execution), and in ZF+AFA it remains a set-theoretic identity. What ZP-J adds is the ROLE that identity describes, stated as the field bot_self_mem, and a proof (T-EXEC) that whatever fills that role is $\perp$.

Because every lattice can fill the role trivially, carrying AFAStructure is not evidence that $\perp = \{\perp\}$ holds of a lattice. In ZF+AFA the set $Q = \{Q\}$ exists and is unique, a consequence of the axiom (Aczel 1988, Example 1.3); identifying the framework's $\perp$ with that set is the commitment (ZP-E, Remark R-AFA). The role, stated as a hypothesis where any reader can see it, is what the Lean checks.

III. The Formal Chain

The derivation chain entering ZP-J had one remaining gap: CC-1 was a committed starting point, not a derived one. Sections I–VI restate that commitment in terms of the Quine-atom role, where it becomes an equivalence rather than a bare choice. Sections VII–X extend the chain further: they derive AFAStructure's laws from stronger structures, provide concrete models, and prove decoration uniqueness for finite graphs over abstract DecorationUniverses — an analogue of the uniqueness clause of what AFA asserts as an axiom. Every node is either a proved theorem or a typeclass field:

- ZP-A axioms A1–A4: semilattice structure (ZPSemilattice fields).
- ZP-B: 2-adic topology and irreversibility (proved from Mathlib).
- ZP-C: information theory, L-INF, L-RUN (proved axiom-free or from Mathlib).
- ZP-D: state layer, orthogonality (proved from ZP-A and ZP-B).
- ZP-E: T-SNAP, DA-1, DA-2 (T-SNAP and DA-2 proved axiom-free; DA-1 Path 3 outside Lean scope).
- ZP-J: AFAStructure fields (selfMem, quine_unique, bot_self_mem). T-EXEC, J1, CC-1: proved axiom-free. Sections VII–X (Aczel connection, abstraction chain, concrete instances, decoration uniqueness): all sorry-free. ✓

DA-1 Path 3 (the AIT/Kolmogorov complexity argument in ZP-E) remains outside Lean scope for the same reason it always has: Kolmogorov complexity is uncomputable and absent from Mathlib. ZP-J does not affect DA-1's status.

Section VII: The Aczel Connection — What the Encoding Does Not Settle

Formalised in `AczelConn.lean` as an extension of the T-EXEC typeclass encoding. The results are choice-free statements about a lattice predicate; they do not reach Aczel's set-theoretic question.

I. Aczel's Use of Dependent Choice

Aczel (Non-Well-Founded Sets, 1988, ch. 6, Theorem 6.5) proves that for a set-continuous operator Φ , $J\Phi = \bigcup \{x \mid x \subseteq \Phi x\}$ is the largest fixed point of Φ , and that every class X with $X \subseteq \Phi X$ is contained in $J\Phi$ (part 2). In the proof of part (2) he uses the axiom of Dependent Choice (DC) to build an infinite sequence of subsets of X , and notes: "I do not know if this use of the axiom of dependent choices was essential."

The step that uses DC passes from an element a of a CLASS X with $X \subseteq \Phi X$ to a SET x with $a \in x \subseteq \Phi x$, which is needed because $J\Phi$ is a union of sets. That step is where Aczel's question lives, and it is a question about a proof in set theory.

II. The `J_self` Set and Its Structure

In ZP's encoding the lattice-level counterpart is $J_self = \{x : L \mid selfMem(x)\}$, the set of self-containing elements, defined directly by comprehension on the predicate `selfMem`. No set-continuous operator is defined and `J_self` is not built as a union of sets, so the step that uses DC in Aczel's proof has no counterpart here. The results about `J_self` follow immediately from `AFAstructure` (Lean: `AczelConn.lean § I`):

`J_self` Theorems (`AczelConn.lean § I`)

<code>bot ∈ J_self</code> : the bottom element is self-containing. (<code>AFAstructure.bot_self_mem</code> .)
<code>J_self_eq_bot</code> : every $x \in J_self$ satisfies $x = \perp$. (One step: <code>quine_unique x ⊥ hx bot_self_mem</code> .)
<code>J_self_eq_singleton_bot</code> : $J_self = \{\perp\}$. No <code>Classical.choice</code> (<code>[propext, Quot.sound]</code>). ✓
<code>J_self_is_largest</code> : for any set S of self-containing elements, $S \subseteq J_self$. This holds by the definition of <code>J_self</code> , for any predicate, and uses no uniqueness: it has the shape of Aczel 6.5 part (2) and is not an instance of it. ✓

`quine_unique` is used once, to identify the only element of `J_self` as $\perp$. That identification corresponds to no clause of Theorem 6.5, and `J_self_is_largest` does not use it. The absence of `Classical.choice` is a measured fact about these Lean terms. It does not show that DC is inessential in Aczel's proof, for this or any operator.

III. The Abstract Principle: A Unique Witness Determines Its Set

The identification is not specific to self-membership. It holds for any predicate with a unique witness. `AczelConn.lean` makes this explicit:

Theorem singleton_from_unique_witness (AczelConn.lean § II)

For any type α and predicate $P : \alpha \rightarrow \text{Prop}$: if w satisfies $P(w)$ and $\forall x, P(x) \rightarrow x = w$, then $\{x \mid P(x)\} = \{w\}$.

Proof: pure set extensionality. No Classical.choice ([propext, Quot.sound]). ✓

Application: selfMem_determines_singleton — $\{x : L \mid \text{selfMem}(x)\} = \{\perp\}$ for any AFAStructure lattice L .

Scope note: ZP-J does not answer Aczel's question, for the self-membership case or any other. His question concerns a use of DC inside a proof in set theory; the results here are choice-free statements about a lattice predicate and do not reach that proof. This document takes no position on whether that use of DC is essential.

Section VIII: The Abstraction Chain — ValuationStructure → AbstractSelfApp → AFAStructure

Developed in SelfApp.lean and Scale.lean after T-EXEC was established; the abstraction chain derives AFAStructure's laws from stronger structures: a fixed-point structure and, beneath it, a valuation structure.

I. The Question: Can AFAStructure's Fields Be Derived?

AFAStructure has three fields: selfMem (a predicate), quine_unique (AFA uniqueness), and bot_self_mem (the bridge). These appear as structural prerequisites — typeclass fields rather than freestanding axioms, but still commitments that any instance must supply — two to PROVE, and selfMem to DEFINE, since it is data rather than a law. The abstraction chain asks: can the two laws themselves be derived from other structure?

The answer is yes, in two steps, with selfMem supplied by a definition along the way. First, AbstractSelfApp provides a self-application operation with unique_fp as a law, from which the two AFAStructure LAWS become derived theorems. Then ValuationStructure explains why $\perp$ is the unique fixed point, making even unique_fp a theorem rather than a field.

II. AbstractSelfApp — The Minimal Fixed-Point Structure

AbstractSelfApp encodes the abstract pattern common to both the set-theoretic and 2-adic domains: a self-application operation whose unique fixed point is $\perp$.

AbstractSelfApp Typeclass (SelfApp.lean § I)

class AbstractSelfApp (L : Type*) [ZPSemilattice L] with:

(1) selfApp : L → L — the self-application operation.

(2) fixed_bot : selfApp($\perp$) = $\perp$ — $\perp$ is a fixed point.

(3) unique_fp : $\forall x : L, \text{selfApp}(x) = x \rightarrow x = \perp$ — $\perp$ is the ONLY fixed point.

From the two laws fixed_bot and unique_fp (selfApp is data), the two AFAStructure LAWS become derived theorems, and selfMem is supplied as a definition (selfMemDerived) rather than proved — it is data:

Derived Results from AbstractSelfApp (SelfApp.lean § II-III)

selfMemDerived $x := \text{selfApp}(x) = x$ — self-containment as fixed-point property.

derived_bot_self_mem: selfMemDerived($\perp$) — from fixed_bot. ✓

derived_quine_unique: any two self-containing elements are equal — each equals $\perp$ by unique_fp, so equal to each other. ✓

selfMem_eq_singleton_bot: $\{x \mid \text{selfMemDerived}(x)\} = \{\perp\}$ — via singleton_from_unique_witness. No Classical.choice. ✓

instance toAFAStructure: any AbstractSelfApp gives an AFAStructure. The two laws are theorems, not additional commitments; selfMem is a supplied definition. ✓

III. ValuationStructure — Why $\perp$ is the Unique Fixed Point

AbstractSelfApp assumes unique_fp. ValuationStructure derives it. The key insight is the 2-adic argument: if scale increases valuation by 1 at every non- $\perp$ element, then $\text{scale}(x) = x$ implies $\text{val}(x) = \text{val}(x) + 1$ — impossible for any finite valuation. Only $\perp$, which has infinite valuation ($\text{val}(\perp) = \top$), can be a fixed point.

ValuationStructure Typeclass (Scale.lean § I)

class ValuationStructure (L : Type*) [ZPSemilattice L] with:

(1) scale : L → L — the self-application (scaling) operation.

(2) val : L → $\mathbb{N}_\infty$ — valuation into the extended naturals.

(3) scale_bot : scale($\perp$) = $\perp$ — $\perp$ is a fixed point of scale.

(4) val_bot : val($\perp$) = $\top$ — $\perp$ has infinite valuation.

(5) val_unique : $\forall x, \text{val}(x) = \top \rightarrow x = \perp$ — only $\perp$ has infinite valuation.

(6) val_scale : $\forall x \neq \perp, \text{val}(\text{scale}(x)) = \text{val}(x) + 1$ — scale strictly increases valuation at non- $\perp$ elements.

Derived Results from ValuationStructure (Scale.lean § II-IV)

val_finite_of_ne_bot: $x \neq \perp \Rightarrow \text{val}(x) \neq \top$ — contrapositive of val_unique. ✓

scale_ne_fixed: $x \neq \perp \Rightarrow \text{scale}(x) \neq x$ — $\text{val}(\text{scale}(x)) = \text{val}(x) + 1 \neq \text{val}(x)$ since $\text{val}(x)$ is finite. ✓

scale_unique_fp: scale(x) = x $\Rightarrow$ x = $\perp$ — from scale_ne_fixed. ✓

instance toAbstractSelfApp: selfApp = scale; fixed_bot and unique_fp are theorems. No additional fields. ✓

IV. The Full Chain and the 2-Adic Parallel

The complete derivation chain is:

ValuationStructure $\rightarrow$ AbstractSelfApp $\rightarrow$ AFAStructure Four valuation axioms $\rightarrow$ Two fixed-point fields $\rightarrow$ Three AFA fields At each step, the LAWS of the lower typeclass become derived theorems; the data fields are supplied, not proved.

What this chain does NOT establish, and the Lean says so explicitly. AbstractSelfApp is a STRUCTURE, not a property: it supplies a chosen self-application map, and carrying one says nothing about the carrier, because every ZPSemilattice carries one. The NO-GO gauge in ZeroParadox/Computability/SelfApp.lean §V records the consequence: because toAFAStructure is an instance, the whole AFA layer — IsQuineAtom, bot_is_quine_atom, T-EXEC — comes free from a degenerate map, so the anti-foundation layer is satisfied by a structure in which nothing refers to anything. The chain is therefore a statement about what FOLLOWS from the structure, not evidence that any particular carrier has it non-trivially. Membership is not an argument; a witness that the framework's real instances differ from the degenerate one is. Each arrow is an instance, so it runs from the stronger hypothesis to the weaker: every ZPSemilattice carries AbstractSelfApp and AFAStructure, while a ValuationStructure on a carrier with any element besides $\perp$ forces that carrier to be infinite (valuationStructure_forces_infinite, ZeroParadox/Valuation/ScaleBridge.lean § VI). The chain explains the AFA laws; it does not reduce what must be assumed.

At each level of the chain, the 2-adic parallel holds. At the AbstractSelfApp level: multiplication by 2 in $\mathbb{Q}_2$ has 0 as its unique fixed point ($2x = x \Rightarrow x = 0$, by ring arithmetic), and $\{x \in \mathbb{Q}_2 \mid 2x = x\} = \{0\}$ by singleton_from_unique_witness. At the ValuationStructure level: in $\mathbb{Z}_2$, scale = $\times 2$ and val = 2-adic valuation satisfy all four axioms — in particular, $v_2(2x) = v_2(x) + 1$ for $x \neq 0$ (proved from PadicInt.valuation_mul and PadicInt.valuation_p in Mathlib). No ZPSemilattice or ValuationStructure instance on $\mathbb{Z}_2$ is registered, so the parallel is proved as standalone theorems demonstrating the same proof structure closes both cases. That is a fact about what is declared, not what is possible: ZeroParadox/Valuation/Scale.lean § V proves $\mathbb{Z}_2$ admits a ZPSemilattice carrying a ValuationStructure.

Section IX: Concrete Instances — $\mathbb{N}_\infty$ and OntologicalStates

Developed in Model.lean and OntBridge.lean to ground the abstraction chain in concrete types; OntologicalStates takes a shorter path because a two-element carrier admits no ValuationStructure.

I. $\mathbb{N}_\infty$ — The Canonical ValuationStructure Instance

$\mathbb{N}_\infty = \text{WithTop } \mathbb{N}$ (the extended naturals) carries a ZPSemilattice with join = min and bot = $\top$ (the natural maximum). The ZP partial order reverses $\mathbb{N}_\infty$'s natural order: $x \leq y$ iff min $x y = y$ iff $x \geq y$. So $\top$ is the ZP-bottom (valuation ∞ , unique fixed point) and 0 is the ZP-maximum (fully constrained).

$\mathbb{N}_\infty$ Instances (Model.lean)

instNatInfZPS: ZPSemilattice $\mathbb{N}_\infty$ with join = min, bot = $\top$. A1–A3: min is associative, commutative, idempotent. A4: min $\top x = x$ because $\top$ is the maximum. ✓

instNatInfVal: ValuationStructure $\mathbb{N}_\infty$ with scale = $(\cdot + 1)$, val = id. scale_bot: $\top + 1 = \top$ (WithTop.top_add). val_bot: id $\top = \top$. val_unique: id $x = \top \Rightarrow x = \top$. val_scale: $x \neq \top \Rightarrow \text{id}(x + 1) = \text{id}(x) + 1$ (rfl). ✓

Derived AFA Content on $\mathbb{N}_\infty$ (Model.lean § III)

natInf_scale_unique_fp: $x + 1 = x \Rightarrow x = \top$. (The unique fixed point of $(\cdot + 1)$ in $\mathbb{N}_\infty$ is $\top$.) ✓

natInf_selfMem_singleton: $\{x : \mathbb{N}_\infty \mid x + 1 = x\} = \{\top\}$. ✓

Via toAbstractSelfApp and toAFAStructure, $\mathbb{N}_\infty$ carries a full AFAStructure with the two laws as theorems and selfMem supplied as a definition. ✓

II. OntologicalStates — The Direct AbstractSelfApp Path

OntologicalStates = {null, exist} (ZP-B's two-element state space) cannot follow the ValuationStructure path: a ValuationStructure on a carrier with any element besides $\perp$ forces that carrier to be infinite (valuationStructure_forces_infinite), so a two-element type admits none. Instead it takes the direct path to AbstractSelfApp.

The self-application operation is the constant-to-null function: every element maps to null. null is the unique fixed point because $\text{null} \mapsto \text{null}$ (fixed_bot) and $\text{exist} \mapsto \text{null} \neq \text{exist}$ (unique_fp holds vacuously for exist). AFA content follows immediately from the AbstractSelfApp instance.

OntologicalStates Instances (OntBridge.lean)

instOntZPS: ZPSemilattice OntologicalStates with null-identity join and bot = null. All four axioms proved by case analysis. ✓

instOntSelfApp: AbstractSelfApp OntologicalStates with selfApp = constant-to-null. fixed_bot: $\text{null} \mapsto \text{null} = \text{null}$ (rfl). unique_fp: $\text{null} \mapsto \text{rfl}$; $\text{exist} \mapsto \text{absurd hx}$ (by decide). ✓

Derived AFA Content on OntologicalStates (OntBridge.lean § III)

ont_bot_self_mem: null is self-containing. ✓

ont_quine_unique: any two self-containing elements of OntologicalStates are equal. ✓

ont_selfMem_singleton: $\{x \mid \text{selfMemDerived}(x)\} = \{\text{null}\}$. No Classical.choice. ✓

Two concrete instances, two paths through the chain. $\mathbb{N}_\infty$ takes the full ValuationStructure route. OntologicalStates bypasses ValuationStructure and connects directly to AbstractSelfApp. Both deliver the same conclusion: the unique self-containing element is the bottom. The architecture is sound because each type takes the path the mathematics allows.

Section X: APG Decoration Uniqueness

Formalised in APG.lean as the most recent addition; this section proves decoration uniqueness for finite accessible pointed graphs over abstract DecorationUniverses — an analogue of AFA's uniqueness clause, not that clause and not AFA itself — using the typeclass chain established in Sections VIII–IX.

I. Accessible Pointed Graphs and Decorations

An Accessible Pointed Graph (APG) is a directed graph (Quiver) with a distinguished root vertex from which every vertex is reachable via directed paths. AFA is an axiom, not a theorem: every graph has a unique decoration — a labelling of vertices by sets satisfying the membership equation $d(v) = \{d(w) \mid v \rightarrow w\}$ (Aczel 1988, ch. 1 p. 6). Its well-founded counterpart, Mostowski's Collapsing Lemma (p. 4), is a result: every well-founded graph has a unique decoration.

ZP-J's uniqueness result is stated for an abstract DecorationUniverse rather than sets. This avoids ZFSet (which satisfies Foundation, making self-loops impossible) and places the result in the typeclass framework established above.

Typeclass: DecorationUniverse (APG.lean § II)

```
class DecorationUniverse (U : Type*) [ZPSemilattice U]
```

```
[ValuationStructure U] where
```

```
collect : Set U → U
```

```
collect_singleton : ∀ x : U, collect {x} = scale x
```

```
collect_val_ge : ∀ (S : Set U) (x : U), x ∈ S → val (collect S) ≥ val x + 1
```

Exactly two laws. They pin the SINGLETON case and a lower bound; they do not require collect to assemble a parent value from its children.

A valid decoration $d : V \rightarrow U$ satisfies $d(v) = \text{collect}(\{d(w) \mid v \rightarrow w\})$ at every vertex v .

II. The Valuation Argument for Cyclic Vertices

The first key results handle cyclic vertices. The argument is a chain of two inequalities, not an iterated equation. Let w be the next vertex on the cycle and p the path from w back to v . Because w is a child of v and $d(v)$ collects over d of v 's children, `collect_val_ge` gives $\text{val}(d(v)) \geq \text{val}(d(w)) + 1$. Following p , the lemma `path_val_chain` (private to APG.lean) gives $\text{val}(d(w)) \geq \text{val}(d(v)) + \text{length}(p)$. Together these force $\text{val}(d(v))$ to exceed itself, which no finite value in $\mathbb{N}_\infty$ can do, so $\text{val}(d(v)) = \top$ (infinite) and therefore $d(v) = \perp$. No assumption is made about how many children a vertex has.

The iterated form $d(v) = \text{scale}^k(d(v))$ is not how this proof proceeds. The class laws relate collect to scale only through `collect_singleton`, so composing decoration equations yields that form when each vertex's set of child decorations on the cycle is a singleton — for example, when each vertex has exactly one child. The scale^k lemmas below are genuine results about periodic points of scale, and `kCycle_node_eq_bot` takes that equation as a hypothesis rather than deriving it.

Cyclic Vertex Theorems (APG.lean §§ III-VII')

```
val_iterate: val(scalek(x)) = val(x) + k for x ≠ ⊥. ✓
```

```
scale_iterate_unique_fp: scalek(x) = x ⇒ x = ⊥ for k ≥ 1. ✓
```

Cyclic Vertex Theorems (APG.lean §§ III-VII')

pureSelfLoop_decoration_eq_bot: any valid decoration assigns $\perp$ to a pure self-loop vertex. ✓

kCycle_node_eq_bot: if $d(v) = \text{scale}^k(d(v))$ for some $k \geq 1$ under any valid decoration, then $d(v) = \perp$. The cycle equation is a hypothesis, not derived. ✓

cyclic_decoration_eq_bot: any vertex with a directed cycle through itself receives $\perp$ under any valid decoration. Proof route: collect_val_ge and path_val_chain, with val_finite_of_ne_bot; it uses neither val_iterate nor scale_iterate_unique_fp. ✓

III. The Acyclic Case and the "Infinite Period"

Cyclic vertices have a finite period k — the path returns in k steps, and a valuation bound around the cycle forces $\perp$. Acyclic vertices have an "infinite period" — the path never returns to them, so they carry no cycle of their own for that bound to run around. Read as a coincidence of two charts, the finite k of a cycle is the zero side and its absence is the infinity side. The proved half holds of one vertex in both charts at once: a cyclic vertex receives the order-theoretic zero $\perp$, and $\perp$'s valuation is $\top$, infinite depth (cyclic_decoration_eq_bot with val_bot). The implication runs from the cycle to that value only; no value is claimed for an acyclic vertex from its period alone. The pairing is stated for DecorationUniverses: in AFA set theory a graph with a cycle can picture a set other than Q , such as $0^* = \{\emptyset, 0^*\}$ (Aczel 1988, Example 1.5).

The hardness of the acyclic proof is that "infinite period" gives you nothing to induct on. Reach cardinality — the cardinality of $\{w \mid \text{Path } v \ w\}$, the set of vertices reachable from v — serves as a finite proxy. For any child w of an acyclic vertex v , the reach of w is a strict subset of the reach of v (since v cannot be reached from w without creating a cycle). Reach cardinality is therefore strictly decreasing, providing a well-founded measure for the induction.

Acyclic Induction and Global Uniqueness (APG.lean §§ VIII-IX)

acyclic_induction_step: if two valid decorations d_1, d_2 agree on all children of an acyclic vertex v , they agree on v . (The two child image sets are equal by Set.ext, then rewrite — ordinary congruence, consuming no class axiom.) ✓

decoration_unique [Fintype V]: for any finite APG with root r and any two valid decorations d_1, d_2 into any DecorationUniverse, $d_1 = d_2$. Proof: strong induction on $|\{w \mid \text{Path } u \ w\}|$; cyclic case by cyclic_decoration_eq_bot; acyclic case by acyclic_induction_step with IH on children. ✓

decoration_unique is the ZP analogue of the uniqueness clause of AFA (an axiom, not a theorem): for any finite APG, at most one valid decoration exists into any DecorationUniverse. The proof consumes collect_val_ge (never collect_singleton), val_unique through val_finite_of_ne_bot, and [Fintype V] for the acyclic descent — it gives a sufficient condition for decoration uniqueness, not a characterisation. It does not construct a specific AFA model or derive AFA's axioms from ZP's.

Traceability Register — ZP-J

Claim	Grounded In	New assumption?	Status
T-EXEC: Quine atom = $\perp$	AFAStructure.bot_self_mem + the uniqueness half of IsQuineAtom q	None — typeclass fields	Lean: t_exec — axiom-free ✓
J1: Quine join-identity	T-EXEC + ZP-A A4 (bot_join)	None	Lean: j1_quine_join_identity — axiom-free ✓ (was axiom ax_j1 in stub)
CC-1 (Derived): $S_0 = Q \Rightarrow S_0 = \perp$	T-EXEC	None	Lean: cc1_derived — axiom-free ✓ (restates ZP-A's conditional claim; converse by t_exec_iff)
bot_is_quine_atom	AFAStructure.bot_self_mem + quine_unique	None	Lean: bot_is_quine_atom — axiom-free ✓
t_exec_iff: IsQuineAtom $\leftrightarrow = \perp$	T-EXEC + bot_is_quine_atom	None	Lean: t_exec_iff — axiom-free ✓
bot_unique: join-identity is unique	da2_bottom_characterization	None	Lean: bot_unique — axiom-free ✓ (no AFA required)
J_self = $\{\perp\}$ (no Classical.choice)	AFAStructure.quine_unique — one step	None	Lean: J_self_eq_singleton_bot — no Classical.choice ✓
singleton_from_unique_witness	Set extensionality (propext)	None	Lean: singleton_from_unique_witness — [propext, Quot.sound]; no Classical.choice ✓
AbstractSelfApp $\rightarrow$ AFAStructure	AbstractSelfApp.fixed_bot + unique_fp	None	Lean: toAFAStructure — selfMem := selfMemDerived (a definition); bot_self_mem and quine_unique are theorems ✓
ValuationStructure $\rightarrow$ AbstractSelfApp	ValuationStructure.val_scale + val_unique	None	Lean: scale_unique_fp, toAbstractSelfApp — [propext, Classical.choice, Quot.sound], already in each STATEMENT (ZP-J AFA Addendum § V) ✓
$\mathbb{N}_\infty$: ValuationStructure	$\mathbb{N}_\infty$ arithmetic (WithTop.top_add)	None	Lean: instNatInfZPS, instNatInfVal — each [propext, Classical.choice, Quot.sound] ✓
OntologicalStates : AbstractSelfApp	Constant-to-null map; two-element case analysis	None	Lean: instOntSelfApp — axiom-free ✓
cyclic_decoration_eq_bot	collect_val_ge + path_val_chain + val_finite_of_ne_bot	None	Lean: cyclic_decoration_eq_bot — [propext, Classical.choice, Quot.sound] ✓

Claim	Grounded In	New assumption?	Status
decoration_unique [Fintype V]	cyclic_decoration_eq_bot + acyclic_induction_step + Set.ncard	[Fintype V] — finite graph assumption	Lean: decoration_unique — propext, Classical.choice, Quot.sound ✓
AFAStructure.bot_self_ mem	The role that the ZF+AFA identity $\perp = \{\perp\}$ describes	Typeclass field — a hypothesis; every ZPSemilattice meets it trivially	Not a theorem; a hypothesis T-EXEC carries.

Open Items Register — ZP-J

Item	Status	Description
CC-1 (ZP-A) derivability	RESTATED — not forced	cc1_derived: a state sequence that starts at a Quine atom starts at $\perp$, and by t_exec_iff the converse holds, so the two starting conditions are equivalent. No freestanding axiom; the choice of starting point remains, stated through the role.
AX-J1 bridge axiom	CLOSED — J1 derived	The stub version had ax_j1 as a freestanding axiom. The final version derives J1 from T-EXEC + ZP-A A4. Axiom eliminated.
AFAStructure concrete instances	CLOSED — multiple instances	MachinePhase (ZP-K): machinePhaseAFA : AFAStructure MachinePhase. $\aleph_\infty$: instNatInfZPS + instNatInfVal $\rightarrow$ AbstractSelfApp $\rightarrow$ AFA content. OntologicalStates: instOntSelfApp $\rightarrow$ AFA content directly.
Aczel DC question	NOT ADDRESSED	J_self = $\{\perp\}$ is proved with no Classical.choice, and J_self_is_largest holds by definition. Neither is a statement about Aczel's proof of Theorem 6.5 in set theory, so this document takes no position on whether his use of DC is essential.
DA-1 Path 1 formalisation	PARTIAL — APG case proved	DA-1 Path 1 (ZP-E) invokes $\perp = \{\perp\}$ informally. ZP-J T-EXEC formalises the role identification, not the set identity. decoration_unique (§ X) proves uniqueness for finite APGs over abstract DecorationUniverses. The full ZF+AFA set-theoretic bridge — showing the ZP types literally satisfy the ZF+AFA axioms — remains outside Lean scope.
ZP-A CC-1 cross-check	OPEN — editorial	ZP-A's CC-1 box should match this document: cc1_derived gives an equivalence of starting conditions, not a forcing of the starting point.
Formal ZPSemilattice instance for a ValuationStructure type	PARTIAL — one registered	Model.lean registers both instances on $\aleph_\infty$. ZeroParadox/Valuation/Scale.lean § V proves $\mathbb{Z}_2$ admits a ZPSemilattice carrying a ValuationStructure, as an existence statement, without registering an instance; no registered instance on $\mathbb{Z}_2$ was located in ZeroParadox/**/lean on 2026-09-13.

End of ZP-J | Theorem T-EXEC: Executability of Self-Reference | CC-1 restated as an equivalence — no freestanding
axioms | $J_self = \{\perp\}$ with no Classical.choice | Abstraction chain: $ValuationStructure \rightarrow AbstractSelfApp \rightarrow$
AFAStructure | Instances: $\mathbb{N}\infty$, OntologicalStates | decoration_unique: any two valid decorations of a finite APG into a
DecorationUniverse agree | All results sorry-free in Lean 4